MATH1081
DISCRETE MATHEMATICS

Section 1

Sets, functions and sequences

Welcome to Discrete Mathematics! This subject is likely to be very different from any mathematics you have taken before. This is due partly to the topics studied, and partly to the general approach to the subject.

The topics studied include basic aspects of the kind of mathematics which is important for computing and information sciences. A significant feature of the problems faced in these areas is that they are frequently concerned with *discrete structures* rather than continuous functions. Typically, we will be looking at numerical problems involving integers only, rather than the sort of problems involving real numbers which are studied in calculus. (Nevertheless, we will occasionally use techniques from calculus.) We shall also study problems of logic: these are discrete, in the sense that a meaningful mathematical statement can only have two values – true or false – and not a whole range of values. Our final topic will be of importance in the study of communications networks and optimisation problems.

Among our main concerns in MATH1081 will be to say what we are doing carefully and precisely, and to prove that what we say is accurate. If you are used to taking mathematics subjects where "getting the right answer" is all that matters, you may find this hard to get used to. Of course the right answer is still important, but we will also expect you to give detailed and carefully explained solutions.

How do I do well at discrete maths?

- Start working straight away. Since much of the course material is quite different from what you have done at school, you will find it very difficult to catch up if you fall behind.
- Make sure that you learn the definitions of all new terms as soon as possible. Definitions tell you what the words we are using mean: if you don't know them, then you literally do not know what you are talking about…
- …but the definitions by themselves are not enough! You also need to study examples of the concepts defined in order to try and gain an intuitive grasp of the concepts.
- Doing the exercises from the problem sheets will help you to understand the theory and will make future exercises easier. Don't imagine that you have to completely understand the theory *before* starting the exercises. In nearly every problem there will be *something* you can do, even if you don't completely get it.
- When solving problems, take the time to write out your solutions carefully and neatly, giving a clear explanation of what you are doing and why. A quick scribble is not going to get you many marks in tests and exams, even if it does end with the right answer at the bottom of the page.
- Some parts of this course are not all that difficult. Make sure you practise these bits and get them completely mastered so that you don't throw away easy marks.

SETS

Concept. "A **set** is a collection into a whole, A, of definite and separate objects of our intuition or our thought. These objects are called the **elements** of A." (Georg Cantor, translated by P.E.B. Jourdain.)

Note. A set must be **well-defined**, that is, it must be possible in principle (though maybe very difficult in practice) to decide whether any particular "object" is an element of the set or not.

Notation. We write

$$a \in A$$

to indicate that an "object" a is an element of a set A, and

$$a \notin A$$

to indicate that it is not.

Discussion. Are there any issues with the above definition?

A set can be specified by writing a list or description of its elements within curly brackets, e.g.,

$$\begin{aligned} S &= \{\uparrow, \downarrow, \rightarrow\} \\ T &= \{\text{letters of the English alphabet}\} \\ V &= \{1, 3, 5, 7, 9, \ldots\} \end{aligned}$$

Definition. Two sets are **equal** if and only if they have the same elements. **Note**. Order and repetitions are irrelevant. For example

$$\begin{aligned} S &= \{\rightarrow, \uparrow, \downarrow\} \\ &= \{\uparrow, \uparrow, \downarrow, \uparrow, \rightarrow, \downarrow, \downarrow\} \\ T &= \{a, b, c, \ldots, z\} \\ &= \{z, z, y, y, x, \ldots, a, a\} \\ V &= \{\text{positive odd integers}\} \end{aligned}$$

However

$$S \neq \{\rightarrow, \downarrow, \rightarrow, \rightarrow, \downarrow\}\ .$$

Note. A set does not equal its elements, even if there is only one element:

$$\{a\} \neq a \quad , \quad \{a, b\} \neq a, b\ .$$

Exercise. The following are written in "set-type" notation. Are they genuine sets? Why?

$$\begin{aligned} W &= \{\,\text{large Australian cities}\,\} \\ X &= \{\,\text{real solutions of } x^7 - 5x^4 + x = 1\,\} \\ Y &= \{\,\text{honest politicians}\,\} \\ Z &= \{\,\text{cats with ten legs}\,\} \end{aligned}$$

Z is called the **empty set**, denoted by $\varnothing$ or $\{\ \}$:

$$Z = \varnothing = \{\ \}\ .$$

Definition. Let S be a set. If there are exactly n distinct elements in S, where n is a non-negative integer, we say that S is a **finite** set and that n is the number of elements or **cardinality** of S. The cardinality of S is denoted by $|S|$. A set is said to be **infinite** if it is not finite.

Example. For the sets on previous pages we have

$$|S| = 3 \ , \quad |T| = 26 \ , \quad |X| = 1 \ , \quad |Z| = 0$$

and V is an infinite set.

Exercise. State whether the following sets are finite or infinite; if they are finite, give their cardinality.

- $\{$ integers (strictly) between -2 and 5 $\}$;
- $\{$ real numbers between -2 and 5 $\}$;
- $\{\,\{\,\{\,\{\,\{\,a, b, c, \ldots, z\,\}\,\}\,\}\,\}\,\}$;
- $\{\,a, \{\,a, \{\,a\,\}\,\}\,\}$;
- $\{$ prime numbers $\}$.

The elements of a set may themselves be sets, for example,

$$A = \{\,\{\,1\,\}\,\}$$
$$B = \{\,1,\ \{\,1\,\}\,\}$$
$$C = \{\,\text{spheres in 3-dimensional space}\,\}$$
$$D = \{\,\varnothing\,\}$$

Note.

- $A \neq \{\,1\,\}$, and $A \neq 1$.
- $B \neq A$ since B has two elements, one of which is a set, $\{\,1\,\}$, and the other is not, 1.
- the elements of C are sets since a sphere is a set of points.
- $D \neq \varnothing$, since D has one element (namely, $\varnothing$), while $\varnothing$ has none.

Notation.

$$
\begin{aligned}
\mathbb{N} &= \{\,\text{natural numbers}\,\} \\
&= \{\,0, 1, 2, 3, \ldots\,\} \\
\mathbb{Z} &= \{\,\text{integers}\,\} \\
&= \{\,\ldots, -2, -1,\ 0,\ 1,\ 2, \ldots\,\} \\
\mathbb{Z}^+ &= \{\,\text{positive integers}\,\} \\
&= \{\,1, 2, 3, \ldots\,\} \\
\mathbb{Q} &= \{\,\text{rational numbers}\,\} \\
\mathbb{R} &= \{\,\text{real numbers}\,\} \\
\mathbb{C} &= \{\,\text{complex numbers}\,\}
\end{aligned}
$$

A **universal set** is a set containing all elements of all sets that we will need to talk about in some particular topic. We will often denote a universal set by $\mathcal{U}$. For instance

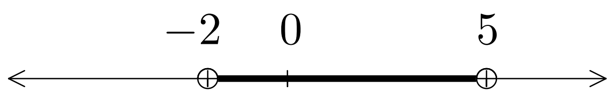
- in calculus the universal set might be $\mathbb{R}$;
- in MATH1131 Algebra Chapter 3 you could take $\mathcal{U} = \mathbb{C}$;
- in geometry, $\mathcal{U} = \{\text{ points in a plane }\}$ or maybe $\{\text{ points in 3–dimensional space }\}$.

Another useful notation is

$$S = \{\, x \in \mathcal{U} \mid \ \ldots \ \} \,,$$

where $\ldots$ denotes some property. The set S then consists of all objects which are in $\mathcal{U}$ and have this property. For example,

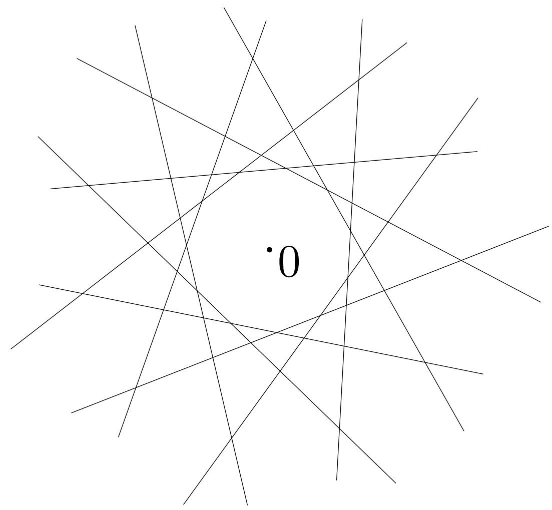
$$\begin{aligned}
S &= \{\, x \in \mathbb{Z} \mid -2 < x < 5 \,\} \\
&= \{\, -1, 0, 1, 2, 3, 4 \,\} \\
T &= \{\, x \in \mathbb{Z} \mid 5 < x < -2 \,\} \\
&= \varnothing \qquad\qquad\qquad\qquad \textbf{not recommended!} \\
V &= \{\, x \in \mathbb{Z} \mid 5 < x \ \text{ or } \ x < -2 \,\} \\
&= \{\, \ldots, -5, -4, -3, 6, 7, 8, \ldots \,\} \\
&\neq T \\
W &= \{\, x \in \mathbb{R} \mid -2 < x < 5 \,\}
\end{aligned}$$



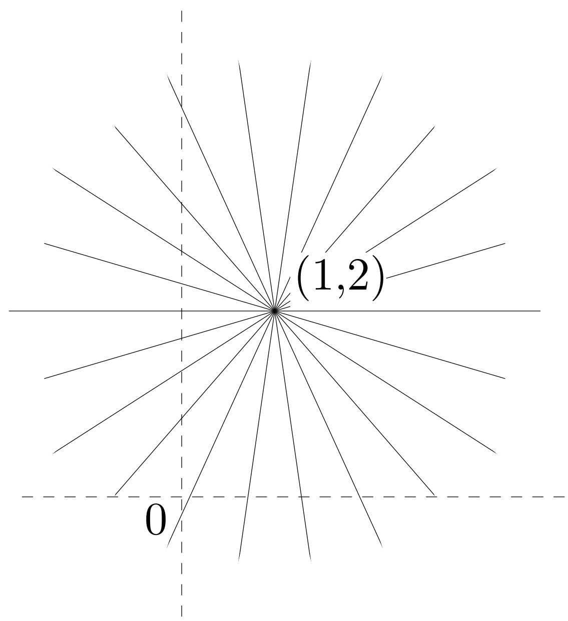
$$\begin{aligned}
X &= \{\, n \in \mathbb{N} \mid n = 2k + 1 \text{ for some } k \in \mathbb{N} \,\} \\
&= \{\, 1, 3, 5, 7, \ldots \,\} \\
&= \{\, \text{odd positive integers} \,\} \\
&= \{\, 2k + 1 \mid k \in \mathbb{N} \,\} \\
\mathbb{Q} &= \left\{ \, \frac{p}{q} \,\middle|\, p, q \in \mathbb{Z}, \ q \neq 0 \right\}
\end{aligned}$$

More examples: let $\mathcal{U} = \{\,\text{straight lines in a plane}\,\}$.

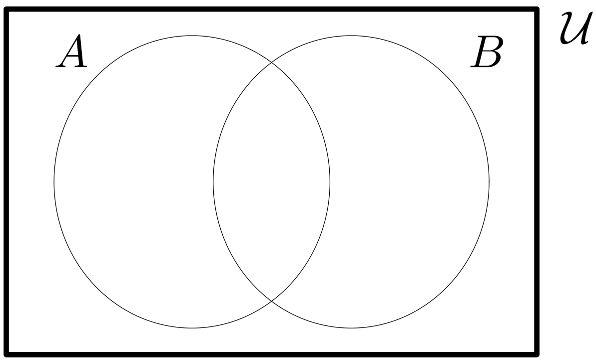
$$Y = \{\, \ell \in \mathcal{U} \mid \text{the shortest distance from } \ell \text{ to the origin is } 2 \,\}$$



$$Z = \{\, \ell \in \mathcal{U} \mid \text{the point } (1,2) \text{ is on } \ell \,\}$$



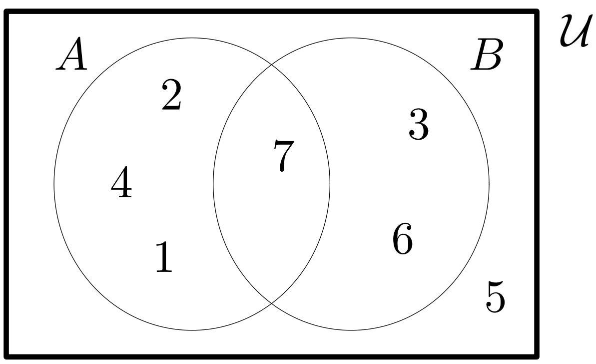
Venn diagrams. In a Venn diagram, sets are depicted as regions in a plane. For example, two sets A and B, subsets of a universal set $\mathcal{U}$, can be drawn thus.



The elements could be shown in the diagram too. For example, if

$$\mathcal{U} = \{\, 1, 2, 3, 4, 5, 6, 7 \,\} \ ,$$
$$A = \{\, 1, 2, 4, 7 \,\} \ , \quad B = \{\, 3, 6, 7 \,\} \ ,$$

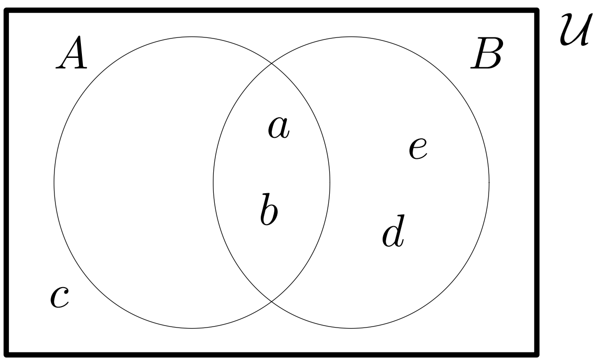
we could draw $\mathcal{U}, A$ and B thus.



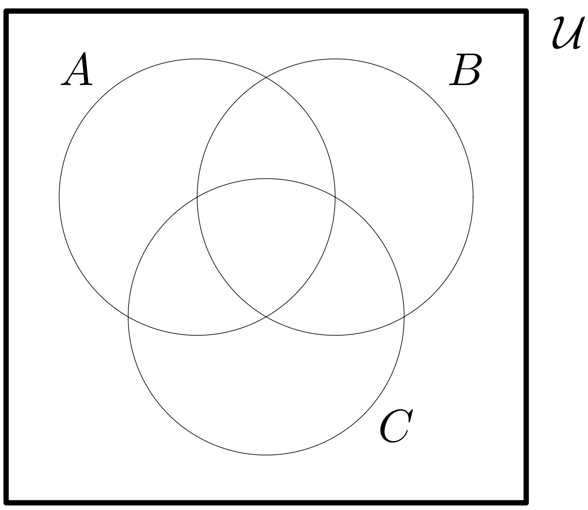
Some regions of a Venn diagram may be considered as empty, so two sets can always be represented as above. For example let

$$\mathcal{U} = \{ a, b, c, d, e \} , \quad A = \{ a, b \} , \quad B = \{ a, b, d, e \} .$$

Here the part of A outside B is empty.



A Venn diagram for three subsets of $\mathcal{U}$ can be drawn thus.



Beyond three sets a Venn diagram is usually too messy for practical use.

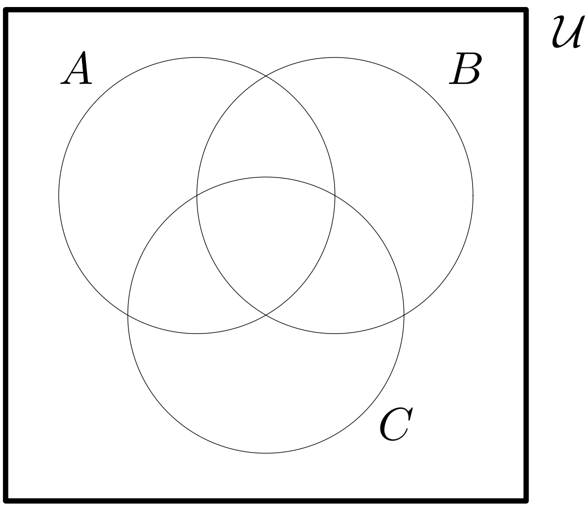
Venn diagrams can be used to solve counting problems such as the following.

Problem. In a survey of 31 people active in sports it was found that 9 participate in athletics, basketball and cricket; 13 participate in athletics and basketball; 14 participate in athletics and cricket; 11 participate in basketball and cricket; 19 participate in athletics; 16 participate in basketball; 22 participate in cricket. How many participate in *exactly one* of these three sports?

Solution. We let $\mathcal{U}$ be the set of all those surveyed, and we define three subsets

$$\begin{aligned} A &= \{\text{ those who participate in athletics }\} \\ B &= \{\text{ those who participate in basketball }\} \\ C &= \{\text{ those who participate in cricket }\} \end{aligned}$$

We can then draw a diagram and write in each section the number of people who play the corresponding combination of sports.



The number who participate in exactly one of the sports is

In a certain village there is a male barber who shaves all the men in the village who do not shave themselves, but none of those who do shave themselves. Does the barber shave himself?

It is conceivable that a set might contain **itself** as an element. For example,

$$M = \{\,\text{mathematical concepts}\,\} \ .$$

Consider

$$S = \{\,\text{sets which are not elements of themselves}\,\} \ .$$

Is this really a set? Why?

What are the important similarities between the two questions above?

Are there any important differences between the first two questions?

Russell's paradox. Let

$$S = \{\,\text{sets which are not elements of themselves}\,\}\ .$$

This is **not** a genuine set since it is impossible to tell whether or not S is an element of S:

- if $S \in S$, then S is "a set which is an element of itself" and so $S \notin S$; but
- if $S \notin S$, then S is "a set which is not an element of itself" and so $S \in S$.

More symbolically, for any set A,

$$A \in S \qquad \Leftrightarrow \qquad A \notin A\ .$$

But if S were really a set we could substitute $A = S$, giving

$$S \in S \qquad \Leftrightarrow \qquad S \notin S\ ,$$

which is impossible.

Definition. The set A is said to be a **subset** of B if and only if every element of A is also an element of B. We use the notation $A \subseteq B$ to indicate that A is a subset of the set B. Also, A is called a **proper subset** of B, written $A \subset B$, if $A \subseteq B$ but $A \neq B$.

Note.

1. If $A = B$ then $A \subseteq B$. Similarly, if $A \subset B$ then $A \subseteq B$.
2. **Important**. If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

Examples.

$$\{1\} \subseteq \{1,2\}\ ; \quad \text{in fact} \quad \{1\} \subset \{1,2\}$$

$$\{a,b,c\} \subseteq \{c,b,a\} \quad \text{but} \quad \{a,b,c\} \not\subset \{c,b,a\}$$

$$\{\uparrow,\downarrow,\rightarrow\} \not\subseteq \{\uparrow,\downarrow,\leftarrow\} \quad \text{and} \quad \{\uparrow,\downarrow,\leftarrow\} \not\subseteq \{\uparrow,\downarrow,\rightarrow\}$$

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

Note carefully the distinction between $\in$ and $\subseteq$.

Examples. True or false?

- $2 \in \{1, 2, 3\}$;
- $2 \subseteq \{1, 2, 3\}$;
- $a \in \{a, \{a\}\}$;
- $a \subseteq \{a, \{a\}\}$;
- $\{2\} \in \{1, 2, 3\}$;
- $\{2\} \subseteq \{1, 2, 3\}$;
- $\{a\} \in \{a, \{a\}\}$;
- $\{a\} \subseteq \{a, \{a\}\}$.

List all the *elements* and all the *subsets* of $\{a, \{a\}\}$.

Answer. The elements are

$$a\ , \quad \{a\}$$

and the subsets are

$$\varnothing\ , \quad \{a\}\ , \quad \{\{a\}\}\ , \quad \{a, \{a\}\}\ .$$

To prove that a certain set A is a subset of another set B we show that any element of A is also an element of B.

Theorem:

$$\{\, x \in \mathbb{R} \mid \sin x = 1 \,\} \subseteq \{\, x \in \mathbb{R} \mid \cos 2x = -1 \,\} \ .$$

Proof. Let $x \in$ LHS. Then $x \in \mathbb{R}$ and $\sin x = 1$. Hence

$$x = \frac{\pi}{2} + 2k\pi \quad \text{for some integer } k,$$

and therefore

$$\cos 2x = \cos(\pi + 4k\pi) = \cos \pi = -1 \ .$$

Since $x \in \mathbb{R}$ and $\cos 2x = -1$, we have $x \in$ RHS. We have shown that any element of the LHS is also an element of the RHS; therefore LHS $\subseteq$ RHS.

Problem. In the previous theorem, is the LHS a *proper* subset of the RHS? Prove your answer.

Problem. Show that

$$\{\, x \in \mathbb{R} \mid \sin x = \tfrac{1}{\sqrt{3}} \,\} \subset \{\, x \in \mathbb{R} \mid \cos 2x = \tfrac{1}{3} \,\} \ .$$

The most usual way to prove that two sets are equal is to show that each is a subset of the other. **Note**. This means that there are **two** separate things to prove.

Theorem. Let

$$A = \{\, 3m - 1 \mid m \in \mathbb{Z} \,\} \quad \text{and} \quad B = \{\, 3n + 8 \mid n \in \mathbb{Z} \,\} \ .$$

Then $A = B$.

Proof. We show, separately, that $A \subseteq B$ and that $B \subseteq A$.

(i) Let $a \in A$. Then $a = 3m - 1$ for some integer m. Thus

$$a = 3m - 1 = 3(m - 3) + 8 = 3n + 8 \ ,$$

where $n = m - 3$, and so $n \in \mathbb{Z}$. Thus $a \in B$.

(ii) Let $b \in B$. Then for some $n \in \mathbb{Z}$ we have

$$b = 3n + 8 = 3(n + 3) - 1 = 3m - 1 \ ,$$

where $m = n + 3$. Since b can be written as $3m - 1$, where $m \in \mathbb{Z}$, we have $b \in A$.

We have shown (i) $A \subseteq B$, and (ii) $B \subseteq A$. Therefore $A = B$.

Exercise. True or false? – suppose that

$$A = \{\, 3m - 1 \mid m \in \mathbb{Z} \,\} \quad \text{and} \quad B = \{\, 3m + 8 \mid m \in \mathbb{Z} \,\} \ .$$

Since $3m - 1$ is never equal to $3m + 8$, we have $A \neq B$.

Some similarities between $\subseteq$ and $\leq$.

- For any set A we have $A \subseteq A$.
- If $A \subseteq B$ and $B \subseteq A$, then $A = B$.
- If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Some differences.

- For any real numbers a, b, either $a \leq b$ or $b \leq a$.
- There is a "smallest" set $\varnothing$, that is, $\varnothing \subseteq A$ for *every* set A.
- In many problems there will be a "largest" set $\mathcal{U}$, that is, $A \subseteq \mathcal{U}$ for *every* set A.
- If a and b are real numbers with $a < b$, there is a real number c between them. Symbolically,

$$\text{if } a < b \text{ then there exists } c \text{ such that } a < c < b.$$

Definition. Given a set S, the **power set** of S is the set of all subsets of the set S. The power set of S is denoted by $\mathcal{P}(S)$.

Examples.

$$\begin{aligned}
\mathcal{P}\big(\{\uparrow,\downarrow\}\big) &= \big\{\varnothing,\{\uparrow\},\{\downarrow\},\{\uparrow,\downarrow\}\big\} \\
\mathcal{P}\big(\{a,\{a\}\}\big) &= \big\{\varnothing,\{a\},\{\{a\}\},\{a,\{a\}\}\big\} \\
\mathcal{P}\big(\{0,1,2\}\big) &= \big\{\varnothing,\{0\},\{1\},\{2\},\{0,1\}, \\
&\qquad\qquad \{0,2\},\{1,2\},\{0,1,2\}\big\} \\
\mathcal{P}\Big(\mathcal{P}\big(\{\uparrow,\downarrow\}\big)\Big) &= \mathcal{P}\Big(\big\{\varnothing,\{\uparrow\},\{\downarrow\},\{\uparrow,\downarrow\}\big\}\Big) \\
&= \big\{\varnothing,\ \{\varnothing\},\ \{\{\uparrow\}\},\ \{\{\downarrow\}\},\ \{\{\uparrow,\downarrow\}\}, \\
&\quad \{\varnothing,\{\uparrow\}\},\ \{\varnothing,\{\downarrow\}\},\ \{\varnothing,\{\uparrow,\downarrow\}\}, \\
&\quad \{\{\uparrow\},\{\downarrow\}\},\ \{\{\uparrow\},\{\uparrow,\downarrow\}\},\ \{\{\downarrow\},\{\uparrow,\downarrow\}\}, \\
&\quad \{\varnothing,\{\uparrow\},\{\downarrow\}\},\ \{\varnothing,\{\uparrow\},\{\uparrow,\downarrow\}\}, \\
&\quad \{\varnothing,\{\downarrow\},\{\uparrow,\downarrow\}\},\ \{\{\uparrow\},\{\downarrow\},\{\uparrow,\downarrow\}\}, \\
&\quad \{\varnothing,\{\uparrow\},\{\downarrow\},\{\uparrow,\downarrow\}\}\big\}
\end{aligned}$$

Note. If X is a set, the statements

$$X \in \mathcal{P}(S) \quad \text{and} \quad X \subseteq S$$

mean exactly the same thing!

Theorem. *Cardinality of the power set.* If $|A| = n \in \mathbb{N}$ then $|\mathcal{P}(A)| = 2^n$.

Proof. If $|A| = n$ we can write

$$A = \{ a_1, a_2, \ldots, a_n \} .$$

In selecting a subset of A we can take the n elements one by one and make a choice: include this element, or not. Thus we have two alternatives for each of n choices, and so the number of choices for the whole procedure is

$$\underbrace{2 \times 2 \times \cdots \times 2}_{n \text{ times}} ,$$

that is, 2^n. So $|\mathcal{P}(A)| = 2^n$.

Example. If $|A| = 2$, then

$$\big|\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathcal{P}(A))))\big| = 2^{65536} ,$$

which is a number of 19729 digits!

SET OPERATIONS

Definition. Let A and B be sets.

1. The **union** of the sets A and B, denoted by $A \cup B$, is the set that contains those elements that are either in A or in B, or in both:

$$A \cup B = \{\, x \mid x \in A \text{ or } x \in B \,\} .$$

2. The **intersection** of the sets A and B, denoted by $A \cap B$, is the set containing those elements in both A and B.
3. Two sets are called **disjoint** if their intersection is the empty set.
4. The **difference** of A and B, denoted by $A - B$, is the set containing those elements that are in A but not in B. The difference of A and B is also called the **complement of** B **with respect to** A.
5. Let $\mathcal{U}$ be the universal set. The **complement** of the set A, denoted by A^c, is $\mathcal{U} - A$, the set of all elements of $\mathcal{U}$ which are not in A.

Alternative notations. Some books (and people!) write

$$A \setminus B \quad \text{instead of} \quad A - B$$

and/or

$$\overline{A} \quad \text{instead of} \quad A^c .$$

Examples.

1. Let

$$\mathcal{U} = \{\,1,2,3,4,5\,\}\ ,$$
$$A = \{\,1,3,5\,\}\ , \quad B = \{\,1,2,3\,\}\ .$$

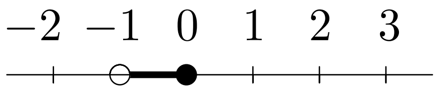
Then

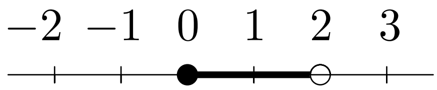
$$A \cup B = \{\,1,2,3,5\,\} = B \cup A$$
$$A \cap B = \{\,1,3\,\} = B \cap A$$
$$A - B = \{\,5\,\}\ , \quad B - A = \{\,2\,\}$$
$$A^c = \{\,2,4\,\}\ , \quad B^c = \{\,4,5\,\}\ .$$

The sets A and B are not disjoint. However the sets $A \cap B$ and A^c are disjoint (exercise: prove that in fact this is true no matter what the sets A and B are).

2. Let $\mathcal{U} = \mathbb{R}$ and consider the subsets

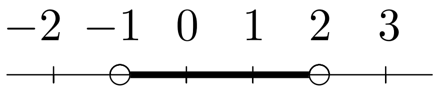
$$I = \{\, x \in \mathbb{R} \mid -1 < x \le 0 \,\}\,, \quad J = \{\, x \in \mathbb{R} \mid 0 \le x < 2 \,\}\,.$$





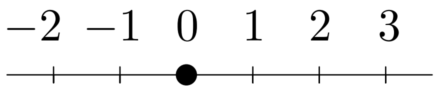
Then we have

$$\begin{aligned} I \cup J &= \{\, x \mid -1 < x \le 0 \text{ or } 0 \le x < 2 \,\} \\ &= \{\, x \mid -1 < x < 2 \,\} \end{aligned}$$



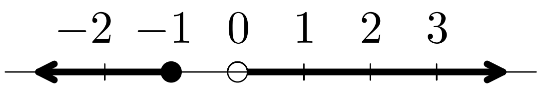
and

$$\begin{aligned} I \cap J &= \{\, x \mid -1 < x \le 0 \text{ and } 0 \le x < 2 \,\} \\ &= \{\, 0 \,\} \end{aligned}$$



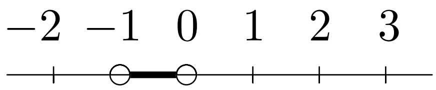
and

$$\begin{aligned} I^c &= \{\, x \mid \text{it is not true that } -1 < x \le 0 \,\} \\ &= \{\, x \mid x \le -1 \text{ or } x > 0 \,\} \end{aligned}$$

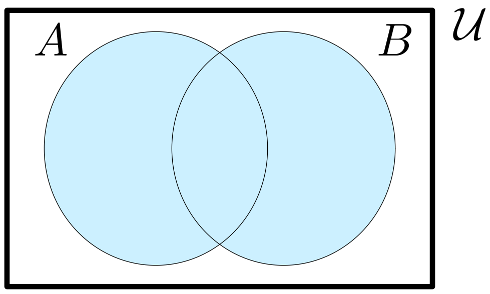


and

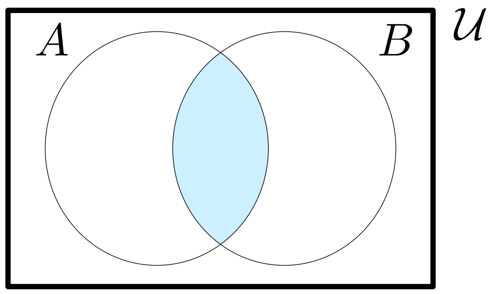
$$I - J = \{\, x \mid -1 < x < 0 \,\}$$



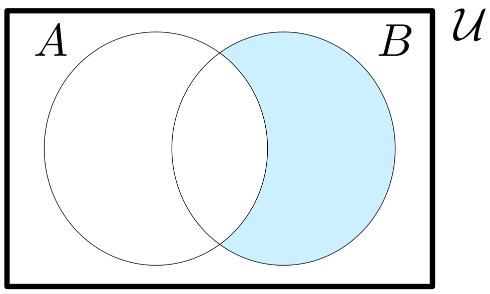
The operations $\cup$, $\cap$, $-$ and ^{c} can be illustrated on Venn diagrams.



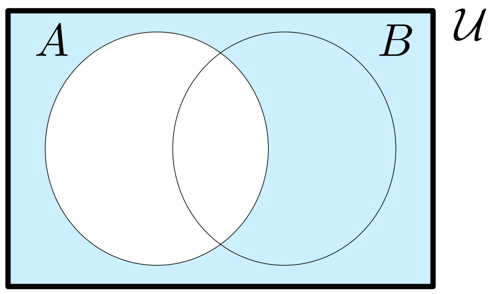
$A \cup B$



$A \cap B$



$B - A$

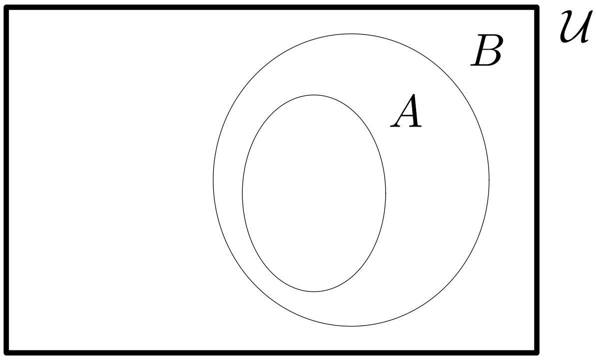


A^c

Sometimes it is useful to draw a Venn diagram in a special way indicating particular knowledge about certain sets.

Exercise. Suppose that $A \subseteq B$. Simplify $A \cup B$, $A \cap B$, and $A - B$.

Solution. Since we know $A \subseteq B$, the Venn diagram can be drawn with A lying entirely inside B.



From the diagram it is easy to see that in this case

$$A \cup B = B \ ,$$
$$A \cap B = A \ ,$$
$$A - B = \varnothing \ .$$

We can also prove the above results algebraically, using the definition of "subset":

$$A \subseteq B \quad \text{means} \quad \text{“if } x \in A \text{ then } x \in B.\text{”}$$

For example, we show that

$$\text{if} \quad A \subseteq B \quad \text{then} \quad A \cap B = A\ .$$

Proof. Suppose that $A \subseteq B$.

First, let $x \in A \cap B$.
    Then by definition $x \in A$ and $x \in B$.
    So $x \in A$.
    Thus $A \cap B \subseteq A$.

On the other hand, let $x \in A$.
    Since $A \subseteq B$ we have also $x \in B$.
    So $x \in A$ and $x \in B$, that is, $x \in A \cap B$.
    Thus $A \subseteq A \cap B$.

If $A \subseteq B$, we have shown that both $A \cap B \subseteq A$ and $A \subseteq A \cap B$, and therefore that $A \cap B = A$.

Exercise. Prove by similar methods that if $A \subseteq B$ then

$$A \cup B = B \quad \text{and} \quad A - B = \varnothing\ .$$

It is often useful to be able to count the number of elements in the union of two sets.

Theorem. *Inclusion–exclusion*. If A and B are finite sets then

$$|A \cup B| = |A| + |B| - |A \cap B| \ .$$

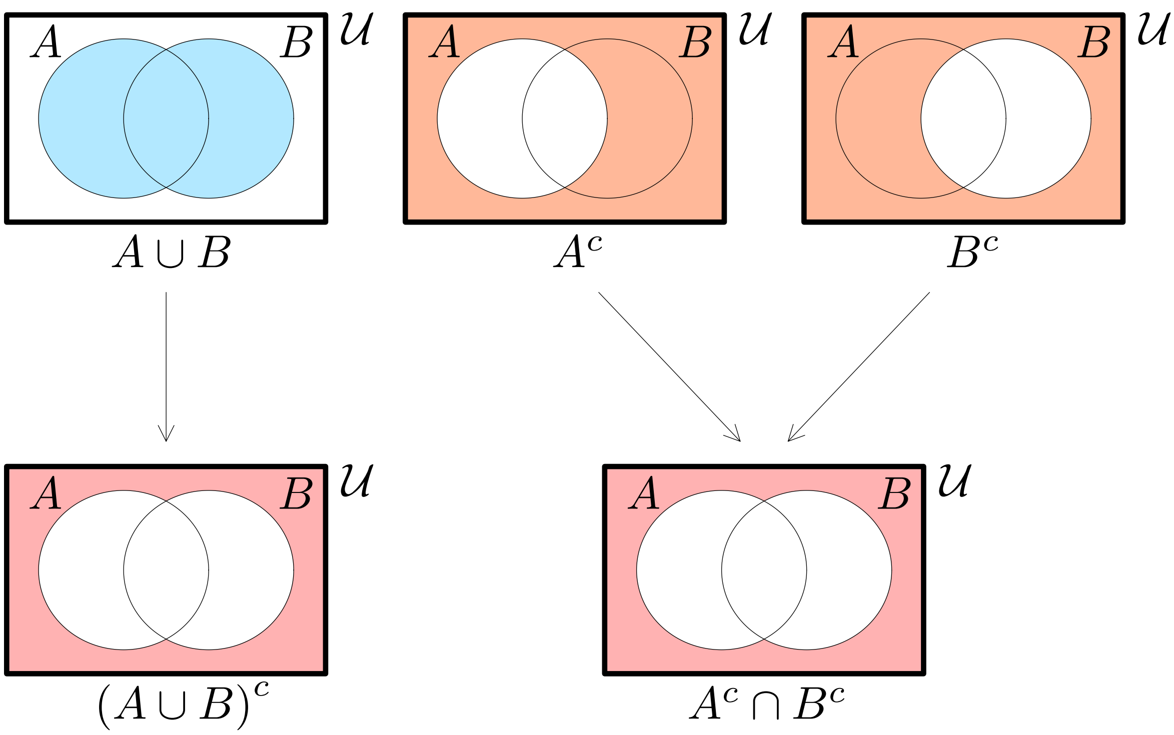
Laws of set algebra. The set operations obey various laws. For example, for any sets A and B we have

$$A \cap B = B \cap A\ .$$

Another example, one of **De Morgan's Laws**:

$$(A \cup B)^c = A^c \cap B^c\ .$$

Proof. An "elementwise" proof is left as an exercise (see also the following example). A Venn diagram proof can be given as follows.



Clearly the two results are the same.

Theorem. *A distributive law*. For any sets A, B and C,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \ .$$

Proof. As usual, we shall show that LHS $\subseteq$ RHS and that RHS $\subseteq$ LHS.

First, let $x \in$ LHS. Then $x \in A$ or $x \in B \cap C$.

Case 1, $x \in A$. In this case $x \in A \cup B$ and $x \in A \cup C$, so $x \in (A \cup B) \cap (A \cup C)$, that is, $x \in$ RHS.

Case 2, $x \notin A$. Then $x \in B \cap C$, so $x \in B$ and $x \in C$. Since $x \in B$ we have $x \in A \cup B$; since $x \in C$ we have $x \in A \cup C$. Thus $x \in (A \cup B) \cap (A \cup C)$, that is, $x \in$ RHS.

This shows that LHS $\subseteq$ RHS.

On the other hand, suppose that $x \in$ RHS. Then $x \in A \cup B$ and $x \in A \cup C$.

Case 1, $x \in A$. Then $x \in A \cup (B \cap C)$.

Case 2, $x \notin A$. But $x \in A \cup B$, so $x \in B$; and $x \in A \cup C$, so $x \in C$. Thus $x \in B \cap C$, so $x \in A \cup (B \cap C)$.

We have now shown that RHS $\subseteq$ LHS, and previously that LHS $\subseteq$ RHS. Therefore LHS = RHS.

BASIC SET IDENTITIES

Let A, B, C be subsets of a universal set $\mathcal{U}$.

1. *Commutative laws.* For all sets A and B,
$$A \cup B = B \cup A \quad \text{and} \quad A \cap B = B \cap A\ .$$

2. *Associative laws.* For all sets A, B and C,
$$\begin{aligned} & A \cup (B \cup C) = (A \cup B) \cup C \\ \text{and} \qquad & A \cap (B \cap C) = (A \cap B) \cap C\ . \end{aligned}$$

3. *Distributive laws.* For all sets A, B and C,
$$\begin{aligned} & A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \\ \text{and} \qquad & A \cup (B \cap C) = (A \cup B) \cap (A \cup C)\ . \end{aligned}$$

4. *Idempotent laws.* For all sets A,
$$A \cup A = A \quad \text{and} \quad A \cap A = A\ .$$

5. *Double complement law.* For all sets A,
$$(A^c)^c = A\ .$$

6. *De Morgan's laws.* For all sets A and B,
$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c\ .$$

7. *Identity laws*. For all sets A,
$$A \cup \varnothing = A \quad \text{and} \quad A \cap \mathcal{U} = A\ .$$

8. *Domination laws*. For all sets A,
$$A \cup \mathcal{U} = \mathcal{U} \quad \text{and} \quad A \cap \varnothing = \varnothing\ .$$

9. *Intersection and union with complement*. For all sets A,
$$A \cap A^c = \varnothing \quad \text{and} \quad A \cup A^c = \mathcal{U}\ .$$

10. *Alternative representation for set difference*. For all sets A and B,
$$A - B = A \cap B^c\ .$$

Question. How on earth am I supposed to remember all those??

Suggestions.

- There are some analogies with the ordinary rules of algebra.
- *Do not* try to mindlessly memorise these rules as strings of symbols, instead do your best to understand what they mean.
- You can cut your work in half by appealing to *duality*.

Duality.

Notice that if we take the above properties (leaving out 10) and replace

$$\begin{array}{cccccc} \cup & \text{by} & \cap\,, & \cap & \text{by} & \cup\,, \\ \varnothing & \text{by} & \mathcal{U}\,, & \mathcal{U} & \text{by} & \varnothing \end{array}$$

we obtain exactly the same properties!

Take any equation that can be proved from these properties and make the above substitutions. The equation we get is called the **dual** of the original equation; it can also be proved from the basic properties and is therefore also true. This is called the **Principle of Duality**:

> A statement of set theory involving no operations except union, intersection and complement is true if and only if its dual is true.

As with ordinary numerical algebra, the basic properties can be used to simplify more complicated expressions.

Example. Simplify $(A^c \cup (B \cap A))^c$.

Solution by algebra. We have

$$
\begin{aligned}
&(A^c \cup (B \cap A))^c \\
&\quad = (A^c)^c \cap (B \cap A)^c && \text{(De Morgan's law)} \\
&\quad = A \cap (B \cap A)^c && \text{(double complement law)} \\
&\quad = A \cap (B^c \cup A^c) && \text{(De Morgan's law)} \\
&\quad = (A \cap B^c) \cup (A \cap A^c) && \text{(distributive law)} \\
&\quad = (A \cap B^c) \cup \varnothing && \text{(intersection with complement)} \\
&\quad = A \cap B^c\ . && \text{(identity law)}
\end{aligned}
$$

Note that having proved this equation, we also have

$$(A^c \cap (B \cup A))^c = A \cup B^c$$

– we get this “for free” by duality. If you want to see it proved from the basic properties, simply dualise each line in the previous proof.

Exercise. By starting with the distributive law, give a different proof of the above result.

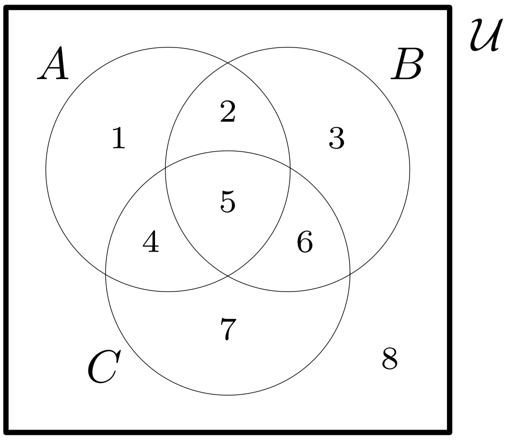
We have seen how to investigate relations between sets using Venn digrams, "elementwise" proofs and the laws of set algebra. Here we look at an example where all three approaches are helpful.

Problem. Decide, with proof, whether or not the equation

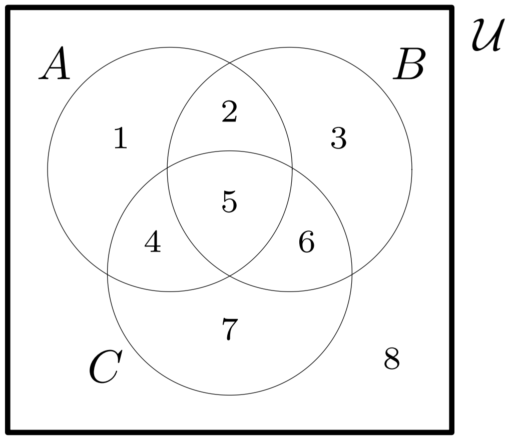
$$(A \cup B) \cap C = A \cup (B \cap C)$$

is always true. If not, find conditions on A, B and C which guarantee that the equation is true.

Solution. We begin by drawing Venn diagrams for the two expressions.



$(A \cup B) \cap C$



$(A \cup B) \cap C$

As the shaded regions are not the same, the two expressions will usually not be equal. To prove this properly, we give a specific example. For instance, take

$$A = \{ 1 \} \,, \quad B = \varnothing \,, \quad C = \varnothing \,;$$

then the sets

$$(A \cup B) \cap C = \varnothing \quad \text{and} \quad A \cup (B \cap C) = \{ 1 \}$$

are not equal.

. . . *continued*

From the diagrams, it seems clear that the two expressions will be equal as long as regions 1 and 2 are empty; this condition is the same as saying that $A \subseteq C$.

Theorem. Let A, B, C be sets. Then

$$(A \cup B) \cap C = A \cup (B \cap C) \tag{$*$}$$

if and only if $A \subseteq C$.

Proof. First, suppose that $A \subseteq C$. Then we have $A \cap C = A$; using the distributive law together with this equation gives

$$(A \cup B) \cap C \;=\; (A \cap C) \cup (B \cap C) \;=\; A \cup (B \cap C)$$

as claimed.

Conversely, suppose that $(*)$ is true. We prove that $A \subseteq C$ by using "elementwise" methods. Let $x \in A$. Then $x \in A \cup (B \cap C)$. From $(*)$ we therefore have $x \in (A \cup B) \cap C$, and so $x \in C$. Therefore $A \subseteq C$, and this completes the proof.

Exercises.

- Another way of expressing $A \subseteq C$ is $A \cup C = C$. Re–do the above proof using this equation instead of $A \cap C = A$.
- It appears from the Venn diagrams that

  $$(A \cup B) \cap C \subseteq A \cup (B \cap C)$$

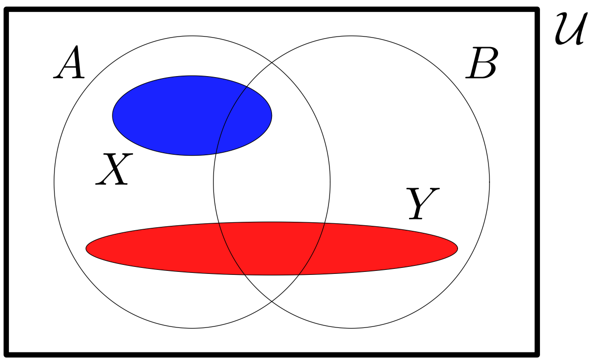
  is always true, no matter whether or not $A \subseteq C$. Prove this using elementwise methods.

Problem. Consider the sets

$$\mathcal{P}(A \cup B) \quad \text{and} \quad \mathcal{P}(A) \cup \mathcal{P}(B)\ .$$

Is the first set always a subset of the second? Or the other way round? Or both? Or neither?

Solution. Once again, before giving a formal answer, we first try to understand the problem by using Venn diagrams. Note that elements of $\mathcal{P}(A)$ are not elements of A but subsets of A; they are therefore represented by regions in the diagram and not by points.



...continued

Answer, part 1. It is not always true that $\mathcal{P}(A \cup B)$ is a subset of $\mathcal{P}(A) \cup \mathcal{P}(B)$. For example, take

$$A = \{ 1, 2 \} , \quad B = \{ 2, 3 \} , \quad Y = \{ 1, 2, 3 \} .$$

Then $Y \subseteq A \cup B$, but $Y \not\subseteq A$ and $Y \not\subseteq B$, that is,

$$Y \in \mathcal{P}(A \cup B) \quad \text{but} \quad Y \notin \mathcal{P}(A) \cup \mathcal{P}(B) .$$

So in this case, $\mathcal{P}(A \cup B)$ is not a subset of $\mathcal{P}(A) \cup \mathcal{P}(B)$.

Answer, part 2. For any sets A and B we have

$$\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B) .$$

Proof. Let $X \in \mathcal{P}(A) \cup \mathcal{P}(B)$; then $X \in \mathcal{P}(A)$ or $X \in \mathcal{P}(B)$; that is, $X \subseteq A$ or $X \subseteq B$.

Suppose that $X \subseteq A$. Since $A \subseteq A \cup B$ we have $X \subseteq A \cup B$, that is, $X \in \mathcal{P}(A \cup B)$. Similarly, if $X \subseteq B$ then $X \in \mathcal{P}(A \cup B)$. So in either case $X \in \mathcal{P}(A \cup B)$, and we have proved that $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$.

Generalised unions and intersections. If $A_1, A_2, \ldots, A_n$ are sets then

$$\bigcup_{i=1}^{n} A_i \quad \text{denotes} \quad A_1 \cup A_2 \cup \cdots \cup A_n$$

$$\text{and} \quad \bigcap_{i=1}^{n} A_i \quad \text{denotes} \quad A_1 \cap A_2 \cap \cdots \cap A_n \ .$$

That is,

$$\bigcup_{i=1}^{n} A_i = \{\, x \mid x \in A_i \text{ for } \textbf{some } i = 1, 2, \ldots \text{ or } n \,\}$$

$$\bigcap_{i=1}^{n} A_i = \{\, x \mid x \in A_i \text{ for } \textbf{all } i = 1, 2, \ldots, n \,\}$$

Examples.

1. Let $A_1 = \{\, a, e, i, o, u \,\}$, $A_2 = \{\, c, f, i, \ldots, x \,\}$ and $A_3 = \{\, n, o, p, q, \ldots, z \,\}$. Then

$$\bigcap_{i=1}^{3} A_i = \{\, o, u \,\}$$

$$\bigcup_{i=1}^{3} A_i = \{\, a, c, e, f, i, l, n, o, p, \ldots, z \,\}$$

2. Let $\mathcal{U} = \mathbb{Z}$ and for any positive integer i write

$$A_i = \{\, i, i+1, i+2, \ldots \,\} \ .$$

Then

$$\bigcup_{i=1}^{n} A_i = \{\, 1, 2, 3, \ldots \,\} = \mathbb{Z}^+$$

$$\bigcap_{i=1}^{n} A_i = \{\, n, n+1, n+2, \ldots \,\} = A_n \ .$$

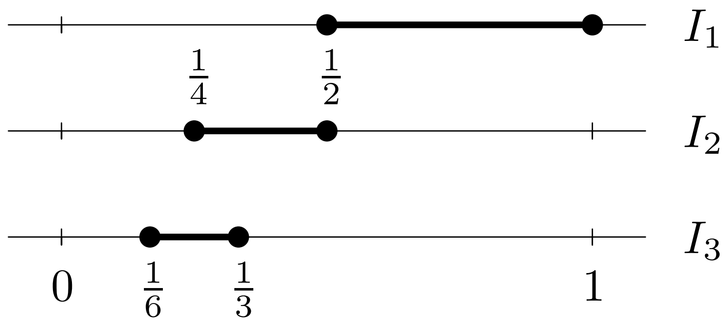
We also write

$$\begin{aligned} \bigcap_{i=1}^{\infty} A_i &= A_1 \cap A_2 \cap A_3 \cap \cdots \\ &= \{\, x \mid x \in A_i \text{ for } \textbf{all } i = 1, 2, 3, \ldots \} \\ &= \varnothing \ . \end{aligned}$$

3. Let $\mathcal{U} = \mathbb{R}$ and

$$I_j = \left\{ x \in \mathbb{R} \;\middle|\; \frac{1}{2j} \le x \le \frac{1}{j} \right\}$$

for $j = 1, 2, 3, \ldots$.



Then

$$\bigcup_{j=1}^{\infty} I_j = \{ x \in \mathbb{R} \mid 0 < x \le 1 \} \, .$$

Proof. First, let $x \in$ LHS. Then

$$x \in I_j \quad \text{for some } j = 1, 2, 3, \ldots ;$$

that is,

$$\frac{1}{2j} \le x \le \frac{1}{j} \quad \text{for some } j .$$

But

$$0 < \frac{1}{2j} \quad \text{and} \quad \frac{1}{j} \le 1 ,$$

so $0 < x \le 1$. Thus $x \in$ RHS.

Conversely, let $x \in$ RHS. Then $\frac{1}{x} \ge 1$. Let j be the largest integer less than or equal to $\frac{1}{x}$. Then $j \ge 1$ and

$$j \le \frac{1}{x} < j + 1 \le j + j ,$$

that is,

$$j \le \frac{1}{x} \le 2j .$$

All quantities here are positive, so we can take reciprocals, reversing the inequalities:

$$\frac{1}{j} \ge x \ge \frac{1}{2j} .$$

Hence $x \in I_j$, so $x \in$ LHS.

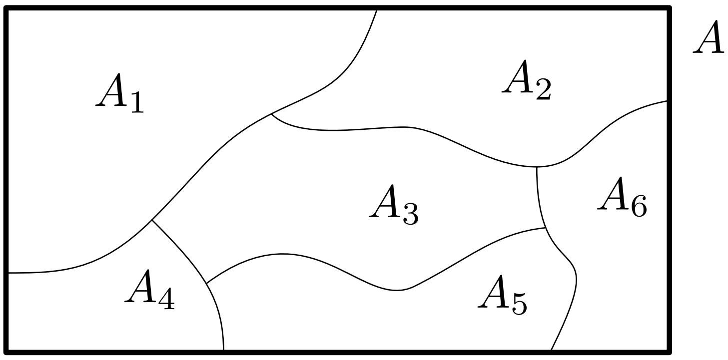
Thus LHS = RHS.

Definition. Sets $A_1, A_2, \ldots, A_n$ are called **disjoint** if and only if they have no elements in common, that is, $A_1 \cap A_2 \cap \cdots \cap A_n = \varnothing$.

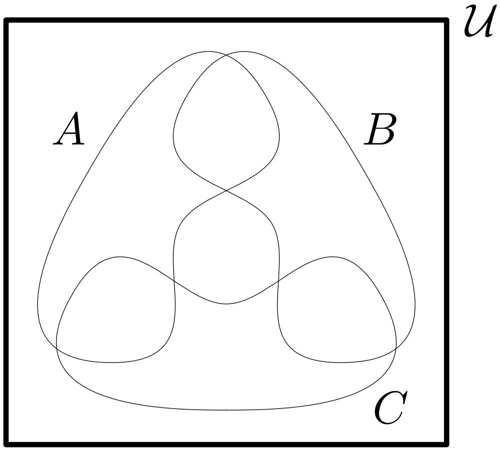
Sets $A_1, A_2, \ldots, A_n$ are **pairwise disjoint** if and only if $A_i \cap A_j = \varnothing$ whenever $i \neq j$.

A set of non-empty sets $\{ A_1, A_2, \ldots, A_n \}$ is a **partition** of a set A if and only if

1. $A = A_1 \cup A_2 \cup \cdots \cup A_n$;
2. $A_1, A_2, \ldots, A_n$ are pairwise disjoint.



The following Venn diagram illustrates three sets which are disjoint, but not pairwise disjoint.



Examples.

1. Let $A = \{1,2,3,4,5,6\}$ and
$$A_1 = \{3,5\}\ ,\ A_2 = \{1,4,6\}\ ,\ A_3 = \{2\}\ .$$
Then $\{A_1, A_2, A_3\}$ is a partition of A.
2. Let $A = \{\,\text{letters of the alphabet}\,\}$ and
$$V = \{\,\text{vowels}\,\}\ ,\quad C = \{\,\text{consonants}\,\}\ .$$
Then $\{V, C\}$ is a partition of A.
3. The set of positive numbers and the set of negative numbers do not form a partition of $\mathbb{R}$.
4. Let $A = [\,0,1\,]$ and write
$$A_1 = [\,0, \tfrac{1}{3}\,]\ ,\ A_2 = [\,\tfrac{1}{3}, \tfrac{2}{3}\,]\ ,\ A_3 = [\,\tfrac{2}{3}, 1\,]\ .$$
Then $\{A_1, A_2, A_3\}$ is not a partition of A.

Note. The number of sets A_i need not be finite.

5. Let $A = \mathbb{R}$, and for each $j \in \mathbb{Z}$,
$$I_j = [\,j - \tfrac{1}{2}, j + \tfrac{1}{2}\,)\ .$$
Then $\{I_j \mid j \in \mathbb{Z}\}$ is a partition of $\mathbb{R}$.
6. Write $P = \{\,\text{points in a plane}\,\}$ and
$$C_x = \{\,\text{points at distance } x \text{ from the origin}\,\}\ .$$
Then $\{C_x \mid x \in \mathbb{R},\ x \geq 0\}$ is a partition of P.

FUNCTIONS

We usually think of a function as a “rule” which defines specific values of one variable in terms of the values of another variable. This is a useful concept, but there are some difficulties with it.

- What exactly is a “rule”?
- Is $f(x) = \sqrt{x}$ a function?
- Are $f(x) = x^2 - 1$ and $g(x) = (x + 1)(x - 1)$ the same function?

Because of issues like this, we need a more accurate idea of what is meant by a function. We don’t, however, want you to entirely abandon the concept of function as a “rule”: keep it in mind, but be aware that it is not altogether precise.

Definition. Given two sets A and B, the **Cartesian product** of A and B, denoted $A \times B$, is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$. We often write A^2 for $A \times A$.

Example. If $A = \{1, 2\}$ and $B = \{1, 2, 3\}$ then

$$\begin{aligned} A \times B &= \{\,(1,1),\ (1,2),\ (1,3),\ (2,1),\ (2,2),\ (2,3)\,\} \\ A^2 &= \{\,(1,1),\ (1,2),\ (2,1),\ (2,2)\,\}\ . \end{aligned}$$

Definition. A **function** f from a set X to a set Y is a subset of $X \times Y$ with the property

> for each $x \in X$ there is **exactly one** ordered pair $(x, y) \in f$.

The notation $f : X \to Y$ means that f is a function from X to Y. The set X is called the **domain** of f, and the set Y is called the **codomain** of f.

Examples.

1. Let $X = Y = \mathbb{R}$ and

$$f = \{ (x, y) \in X \times Y \mid y = x^2 \} .$$

   Then f is a function from X to Y, since for every $x \in X$, the set f contains *exactly one* ordered pair (x, y), namely, the one with $y = x^2$. The domain and codomain of f are both $\mathbb{R}$.

Notation. We will often write $(x, y) \in f$ as

$$y = f(x) ,$$

and will sometimes say that y is the **image of** x **under** f, or that x is a *pre-image* of y. Note that although each x in the domain will have exactly one image, a value of y in the codomain may have one pre-image, more than one, or none at all. For example...

2. Let $X = Y = \mathbb{R}$, and

$$f = \{\, (x, y) \in X \times Y \mid x = y^2 \,\} \ .$$

Here f is not a function from X to Y. It is *not* true that for every $x \in X$ there is exactly one pair $(x, y) \in f$. For example, take $x = 9$; then

$$(9, 3) \in f \quad \text{and} \quad (9, -3) \in f \ .$$

Similarly, for $x = -9$ there is *no* ordered pair $(-9, y) \in f$. We say that f is **not well-defined**.

3. Let $X = \{\, 1, 2, 3 \,\}$, $Y = \mathbb{Z}$. We can define a function $f : X \to Y$ by

$$\begin{aligned} f &= \{\, (x, y) \in X \times Y \mid y = x^x \,\} \\ &= \{\, (x, x^x) \mid x = 1, 2 \text{ or } 3 \,\} \\ &= \{\, (1, 1),\ (2, 4),\ (3, 27) \,\} \ . \end{aligned}$$

4. Define $f : \mathbb{Q} \to \mathbb{Z}$ by the "rule"

$$f\left(\tfrac{p}{q}\right) = q \ .$$

Then f is not a well-defined function since, for example,

$$\left(\tfrac{1}{2}, 2\right) \in f \quad \text{and} \quad \left(\tfrac{1}{2}, 4\right) = \left(\tfrac{2}{4}, 4\right) \in f \ .$$

5. Two functions which will be of importance in future sections of the course are the floor function and the ceiling function. Each of these has domain $\mathbb{R}$ and codomain $\mathbb{Z}$.

   Definition. For any $x \in \mathbb{R}$, the **floor** of x, denoted by $\lfloor x \rfloor$, is the largest integer which is less than or equal to x. The **ceiling** of x, denoted by $\lceil x \rceil$, is the smallest integer which is greater than or equal to x.

   Examples. Some values of the floor and ceiling:

$$\begin{aligned} \lfloor \pi \rfloor &= 3 \ , & \lceil \pi \rceil &= 4 \ , \\ \lfloor 1081 \rfloor &= 1081 \ , & \lceil 1081 \rceil &= 1081 \ , \\ \lfloor -\tfrac{1}{2} \rfloor &= -1 \ , & \lceil -\tfrac{1}{2} \rceil &= 0 \ . \end{aligned}$$

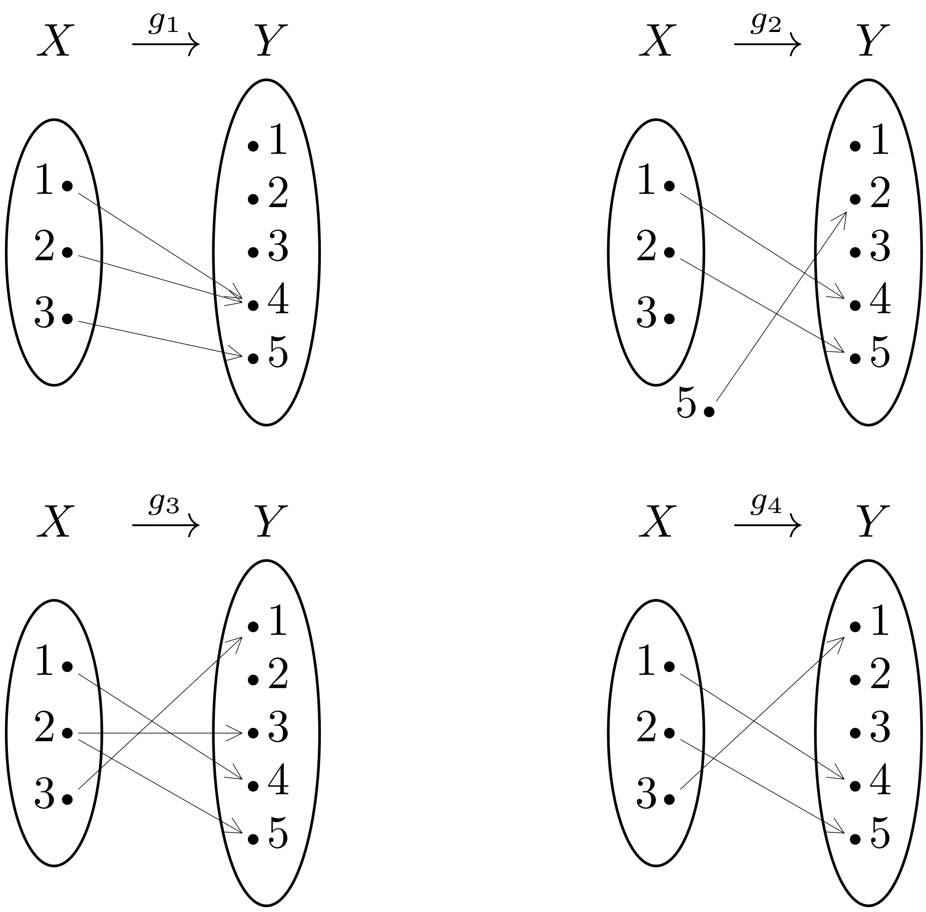
Exercise. Let $X = \{1, 2, 3\}$, $Y = \{1, 2, 3, 4, 5\}$. Which of the following are functions from X to Y?

$$\begin{aligned}
g_1 &= \{ (1,4),\ (2,4),\ (3,5) \} \\
g_2 &= \{ (1,4),\ (2,5),\ (5,2) \} \\
g_3 &= \{ (1,4),\ (2,5),\ (2,3),\ (3,1) \} \\
g_4 &= \{ (1,4),\ (2,5),\ (3,1),\ (1,4),\ (3,1) \}
\end{aligned}$$

Answer.

- g_1 is a function from X to Y;
- g_2 is not since it contains the ordered pair $(5, 2)$, but $5 \notin X$; alternatively, because $3 \in X$ but there is no pair $(3, y) \in g_2$;
- g_3 is not as it contains *two* ordered pairs $(2, 5)$ and $(2, 3)$ with the same first element;
- g_4 is a function from X to Y. **Note** that g_4 is a *set*, so the repetitions of $(1, 4)$ and $(3, 1)$ don't count!

Arrow diagrams. A set of ordered pairs can be illustrated by an arrow diagram.



An arrow diagram represents a function from X to Y if and only if each element of X has exactly one arrow leading away from it into Y (and no arrow starts outside X).

Definition. The **range** of a function $f : X \to Y$ is the set of all $y \in Y$ such that there is an ordered pair (x, y) in f; that is, in symbols,

$$\{ y \in Y \mid (x, y) \in f \text{ for some } x \in X \} ,$$

or

$$\{ f(x) \mid x \in X \} .$$

Examples.

1. The function $f : \mathbb{R} \to \mathbb{R}$ defined by

   $$f = \{ (x, y) \in \mathbb{R}^2 \mid y = x^2 \}$$

   has range

   $$\{ y \mid y \geq 0 \} = \mathbb{R}^+ \cup \{ 0 \} .$$

2. Let $g_1 : \{ 1, 2, 3 \} \to \{ 1, 2, 3, 4, 5 \}$ be given by

   $$g_1 = \{ (1, 4), (2, 4), (3, 5) \} .$$

   Then the range of g_1 is $\{ 4, 5 \}$.

3. The range of the function $\sin : \mathbb{R} \to \mathbb{R}$ is $[\,-1, 1\,]$.

4. The range of the function $h : \mathbb{N} \to \mathbb{N}$ defined by $h(n) = 3n$ is

   $$\{ 3n \mid n \in \mathbb{N} \} = \{ 0, 3, 6, 9, 12, \ldots \} .$$

Note. The codomain of a function, roughly speaking, contains the range together with other, more or less irrelevant elements. The reasons that we bother with the codomain are that it is normally easy to find a codomain, while it may be very difficult to determine the range; and that often we don't need to know the range anyway.

Example (exercise). Suppose that we are interested in a function with "rule" $f(x) = e^x + x^2$, valid for all $x \in \mathbb{R}$. Clearly we will take the domain to be $\mathbb{R}$. Since $x \in \mathbb{R}$ we will always have $f(x) \in \mathbb{R}$, and so we may take the codomain of f to be $\mathbb{R}$. Thus we have a function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f = \{ (x, y) \in \mathbb{R}^2 \mid y = e^x + x^2 \} .$$

But what is the range of f?

Commonly the domain and codomain of a function will be sets of numbers. However, this is not necessary.

Examples.

1. Let A be a finite set, $X = \mathcal{P}(A)$, $Y = \mathbb{N}$. We can define

$$f : \mathcal{P}(A) \to \mathbb{N}, \quad \text{where} \quad f(S) = |S| \ .$$

For example, $f\big(\{\,2, 3, 5\,\}\big) = 3$. As a set of ordered pairs,

$$f = \{\,(S, n) \mid S \text{ has } n \text{ elements}\,\} \ .$$

2. Let $X = \mathbb{R}^+ \times \mathbb{R}^+$, $Y = \mathbb{R}$ and let

$$f = \left\{\,\big((r, h), V\big) \;\Big|\; V = \pi r^2 h\,\right\} \ .$$

3. Write $X = Y = \mathcal{P}(\mathbb{N})$ and

$$f = \Big\{\,(A, B) \;\Big|\; B = A \cap \{\,\text{squares}\,\}\,\Big\} \ .$$

For example

$$f\big(\{\,0, 1, 2, \ldots, 20\,\}\big) = \{\,0, 1, 4, 9, 16\,\} \ ,$$
$$f\big(\{\,\text{cubes}\,\}\big) = \{\,\text{sixth powers}\,\} \ .$$

Definition. A function f is said to be **one-to-one** (or *injective*) if and only if for each y in the codomain there is **only one** ordered pair (x, y) in f (possibly even none at all). In other words, "f is one-to-one" means that for all x_1, x_2 in the domain of f,

$$\text{if } f(x_1) = f(x_2) \text{ then } x_1 = x_2.$$

Example of a function which is not one-to-one. Let $f : \mathbb{Z} \to \mathbb{Z}$ be given by the "rule" $f(x) = x^2$, that is,

$$f = \{ (x, y) \in \mathbb{Z}^2 \mid y = x^2 \} \ .$$

Then f is not one-to-one. For example, let $y = 9$; then there are two ordered pairs (x, y) in f:

$$(3, 9) \in f \quad \text{and} \quad (-3, 9) \in f \ .$$

Similarly, we can use the "$f(x)$" notation by taking $x_1 = 3$, $x_2 = -3$. Then $f(x_1) = f(x_2)$ but $x_1 \neq x_2$, so the statement

$$\text{“if } f(x_1) = f(x_2) \text{ then } x_1 = x_2\text{”}$$

is false.

Examples.

1. Let $f : \mathbb{Z} \to \mathbb{Z}$ be the function

$$f = \{\, (x, y) \in \mathbb{Z}^2 \mid y = 2x + 1 \,\} \ .$$

   Then f is one-to-one. We shall show this by proving

   "if $f(x_1) = f(x_2)$ then $x_1 = x_2$".

   Proof. Let $f(x_1) = f(x_2)$.
   That is, $2x_1 + 1 = 2x_2 + 1$.
   Subtract 1 from both sides, then divide both sides by 2.
   Therefore $x_1 = x_2$.
   Hence f is one-to-one.

2. Let $f : \mathbb{R} \to \mathbb{R}$ be given by the rule

$$f(x) = x^3 + x + 9 \ .$$

   This function is one-to-one. We shall use calculus to prove graphically that

   "if $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$".

   Proof. We have $f(x) = x^3 + x + 9$. Differentiating, $f'(x) = 3x^2 + 1$, which is always positive. Hence the graph $y = f(x)$ has no turning points and is always increasing. Therefore two different x values will give us two different y values.

3. On the other hand, consider

$$f : \mathbb{R} \to \mathbb{R} \quad \text{where} \quad f(x) = x^3 - x + 9 \ .$$

Once again we differentiate,

$$f'(x) = 3x^2 - 1 \ ;$$

it is possible that $f'(x)$ is zero, so f has turning points, and it is clear by drawing the graph and using the *horizontal line test* that f is not one-to-one.

If desired, a specific example of $x_1 \neq x_2$ for which $f(x_1) = f(x_2)$ can conveniently be found by writing $f(x)$ as

$$f(x) = x(x^2 - 1) + 9 \ ;$$

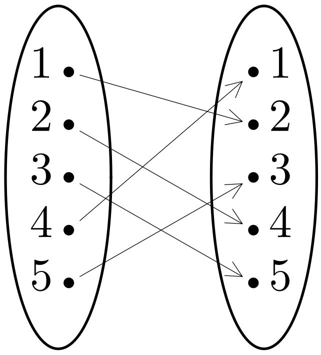
it is then easy to see that $f(1) = f(-1)$.

In terms of arrow diagrams, the requirement for $f : X \to Y$ to be one-to-one is

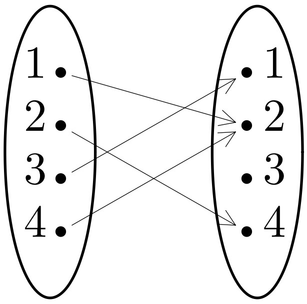
> "any $y \in Y$ has **only one** arrow (maybe none at all) pointing to it".

Examples.

4. The following arrow diagram depicts a one-to-one function



5. ... but this one does not.



$$f = \{\,(1,2),\ (2,4),\ (3,1),\ (4,2)\,\}$$

Definition. A function f from a set X to a set Y is called **onto** (or *surjective*) if and only if for every element $y \in Y$ there is an element $x \in X$ with $y = f(x)$.

In other words, a function is onto if its range equals its codomain.

Examples.

1. Let $f : \mathbb{Z} \to \mathbb{Z}$ be the function

   $$f = \{ (x, y) \in \mathbb{Z}^2 \mid y = 2x + 1 \} .$$

   Then f is not onto. For the range of f is $\{$ odd numbers $\}$, which does not equal the codomain $\mathbb{Z}$. Specifically, if $y = 0$ then there is **no** x such that $y = f(x)$.

2. Let $f : \mathbb{R} \to \mathbb{R}$ be given by the rule

   $$f(x) = x^3 + x + 9 .$$

   From the graph it is seen that for **any** $y \in \mathbb{R}$ there is a value of x such that $y = f(x)$ (that is, $(x, y) \in f$).

   Note. Given y, it will often be difficult to find the actual value of x such that $y = f(x)$. But from the graph this will obviously be true for **some** value of x.

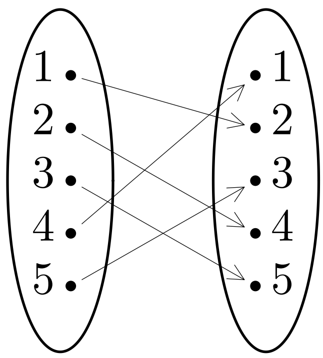
3. The function $f : \mathbb{R} \to \mathbb{R}$ where $f(x) = x^3 - x + 9$ is also onto: from the graph, every y in the codomain is the image of some x in the domain. (Some y values are the image of *more than one* x value, but this is not important.)

In terms of arrow diagrams the requirement for $f : X \to Y$ to be onto is

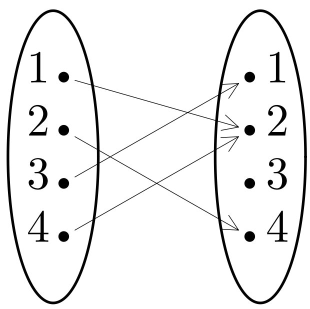
> "any $y \in Y$ has an arrow (possibly more than one) pointing to it".

Examples.

4. The following arrow diagram depicts an onto function



5. ... but this one does not.



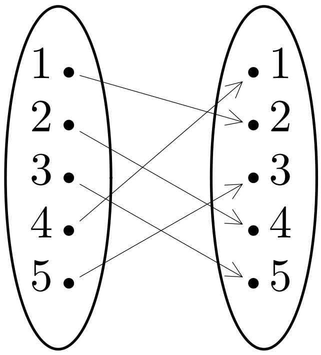
$$f = \{\, (1, 2),\ (2, 4),\ (3, 1),\ (4, 2) \,\}$$

Definition. The function f is a *bijection* if it is both one-to-one and onto.

Examples.

1. Example 2 above shows that the function from $\mathbb{R}$ to $\mathbb{R}$ defined by $f(x) = x^3 + x + 9$ is one-to-one and onto, and is therefore a bijection.

2. Example 4 shows that the function with domain and codomain $\{1, 2, 3, 4, 5\}$ and diagram as follows is a bijection.



The condition in terms of arrow diagrams for $f : X \to Y$ to be a bijection is

"any $y \in Y$ has exactly one arrow pointing to it".

Exercise. We define a function $f : \mathbb{Z} \to \mathbb{Z}$ by the rule

$$f(n) = \begin{cases} n + 5 & \text{if } n \geq 0 \\ n - 2 & \text{if } n < 0. \end{cases}$$

- Is f one-to-one?
- Is f onto?
- Is f a bijection?

Composition of functions.

Definition. Let $g : X \to Y$ and $f : Y \to Z$. Then the **composition** of f and g, denoted by $f \circ g$, is the function from X to Z defined by

$$(f \circ g)(x) = f\big(g(x)\big) \ .$$

Examples.

1. Let $f : \mathbb{Z} \to \mathbb{Z}$ and $g : \mathbb{Z} \to \mathbb{Z}$ be given by

   $$f(n) = n + 1 \ , \quad g(n) = n^2 \ .$$

   Then $f \circ g$ is the function from $\mathbb{Z}$ to $\mathbb{Z}$ defined by

   $$(f \circ g)(n) \ = \ f\big(g(n)\big) \ = \ g(n) + 1 \ = \ n^2 + 1 \ ,$$

   Likewise, $g \circ f : \mathbb{Z} \to \mathbb{Z}$ is given by

   $$(g \circ f)(n) \ = \ g\big(f(n)\big) \ = \ \big(f(n)\big)^2 \ = \ (n+1)^2 \ .$$

   Note that $f \circ g \neq g \circ f$.

2. We define the functions $f : \{1, 2, 3\} \to \{a, b, c\}$ and $g : \{a, b, c, d\} \to \{\blacktriangle, \bigstar, \blacksquare\}$ by

   $$\begin{aligned} f &= \{(1, c),\ (2, b),\ (3, c)\} \\ g &= \{(a, \blacktriangle),\ (b, \blacksquare),\ (c, \bigstar),\ (d, \blacksquare)\} \ . \end{aligned}$$

   Here $f \circ g$ does not exist, as the codomain of g is not equal to the domain of f. However, $g \circ f : \{1, 2, 3\} \to \{\blacktriangle, \bigstar, \blacksquare\}$ is the function

   $$g \circ f = \{(1, \bigstar),\ (2, \blacksquare),\ (3, \bigstar)\} \ .$$

Problem. Let $f : X \to Y$ and $g : Y \to Z$ be functions. Prove that if $g \circ f$ is one-to-one, then f is one-to-one.

Comment. To prove that f is one-to-one, the general procedure will be the same as for example 1 on page 59: the only new feature is that we have to figure out how to use the fact that $g \circ f$ is one-to-one.

Solution. Let x_1 and x_2 be in X and suppose that $f(x_1) = f(x_2)$. Since g is a function, $g\big(f(x_1)\big) = g\big(f(x_2)\big)$, that is,

$$(g \circ f)(x_1) = (g \circ f)(x_2) \ .$$

But since $g \circ f$ is one-to-one, this implies that $x_1 = x_2$.

We have proved that if $f(x_1) = f(x_2)$ then $x_1 = x_2$: thus, f is one-to-one.

Comment. There are many other relations between composition of functions, one-to-one and onto functions: try problem 33 for a couple of examples.

Theorem. *Bijective = invertible*. Let f be a function from X to Y. If f is a bijection, then there is a function $g : Y \to X$ defined as follows: given any $y \in Y$,

$$g(y) \text{ is the element } x \in X \text{ such that } y = f(x).$$

Conversely, if such a function exists, then f is a bijection.

Proof. Firstly, what we have to show is that the above rule for $g(y)$ really does define a function, that is, that for any $y \in Y$ we have specified **exactly one** $x = g(y) \in X$. Now

- f is onto, so for any $y \in Y$ there is **at least one** value of x such that $y = f(x)$;
- f is one-to-one, so if $f(x_1) = y = f(x_2)$, then $x_1 = x_2$. That is, for each y there is **at most one** value of x such that $y = f(x)$.

Thus for any $y \in Y$ the rule defines exactly one $x \in X$.

Conversely, suppose that such a function g exists: we have to prove that f is one-to-one and onto. Let $y \in Y$ and take $x = g(y)$: then by definition $y = f(x)$. Thus every $y \in Y$ is the image of some $x \in X$, and so f is onto.

To show that f is one-to-one, ...

Note.

1. The function g is called the **inverse** (**function**) of f and is denoted f^{-1}.
2. **NB:** $f^{-1}(x) \neq 1/f(x)$.
3. The rule defining the inverse can be written
$$f^{-1}(y) = x \qquad \Leftrightarrow \qquad y = f(x) \ .$$
4. Note that if f^{-1} exists then the domain of f^{-1} equals the codomain of f, and the codomain of f^{-1} equals the domain of f.
5. In terms of ordered pairs, the inverse of
$$f = \{\, (x, y) \in X \times Y \mid \ \ldots \}$$
is
$$f^{-1} = \{\, (y, x) \in Y \times X \mid (x, y) \in f \,\} \ .$$
This truly is a function provided f is a bijection (proof similar to above).
6. In terms of arrow diagrams, the arrow diagram of f^{-1} is just the arrow diagram of f with arrows reversed. Again, this truly is a function if f is a bijection.

Examples.

- Let $f:\mathbb{R}\to\mathbb{R}$ be given by $f(x)=4x-7$. Then

$$\begin{aligned} x=f^{-1}(y) &\Leftrightarrow\quad y=f(x)\\ &\Leftrightarrow\quad y=4x-7\\ &\Leftrightarrow\quad x=\frac{y+7}{4}\ . \end{aligned}$$

So the inverse of f is

$$f^{-1}:\mathbb{R}\to\mathbb{R}\ ,\quad \text{where}\quad f^{-1}(y)=\frac{y+7}{4}\ ,$$

that is,

$$f^{-1}=\left\{\,(y,x)\in\mathbb{R}^2\;\middle|\;x=\frac{y+7}{4}\,\right\}\ .$$

Since the names of the variables are unimportant we may also write

$$f^{-1}=\left\{\,(x,y)\in\mathbb{R}^2\;\middle|\;y=\frac{x+7}{4}\,\right\}\ .$$

Alternatively, this can be done very simply by using ordered pair notation from the start:

$$f=\{\,(x,y)\in\mathbb{R}^2\mid y=4x-7\,\}$$

so

$$\begin{aligned} f^{-1}&=\{\,(y,x)\in\mathbb{R}^2\mid y=4x-7\,\}\\ &=\{\,(x,y)\in\mathbb{R}^2\mid x=4y-7\,\}\\ &=\left\{\,(x,y)\in\mathbb{R}^2\;\middle|\;y=\frac{x+7}{4}\,\right\}\ . \end{aligned}$$

- Does the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = e^x$ have an inverse?

  Answer: no, because f is not onto. **Note** that we are taking a more careful approach than is done in calculus, where we would probably say that this function does have an inverse. In both calculus and discrete, it is correct to say that
  $$g : \mathbb{R} \to \mathbb{R}^+ \quad \text{where} \quad g(x) = e^x$$
  has an inverse.

Composition of a function and its inverse. If we imagine f as a function which "does something" to a variable, then f^{-1} is the function which "undoes" it.

Example. Recall that if

$$f : \mathbb{R} \to \mathbb{R} \;, \quad \text{where} \quad f(x) = 4x - 7 \;,$$

then

$$f^{-1} : \mathbb{R} \to \mathbb{R} \;, \quad \text{where} \quad f^{-1}(y) = \frac{y+7}{4} \;.$$

Note that for any $x \in \mathbb{R}$,

$$\big(f^{-1} \circ f\big)(x) = f^{-1}\big(f(x)\big) = \frac{f(x)+7}{4} = \frac{(4x-7)+7}{4} = x \;.$$

Thus $f^{-1} \circ f$ is the function $\iota : \mathbb{R} \to \mathbb{R}$, which *does nothing* to each $x \in \mathbb{R}$, and is called the **identity** function on $\mathbb{R}$. Similarly, for each $y \in \mathbb{R}$ we have

$$\big(f \circ f^{-1}\big)(y) = f\big(f^{-1}(y)\big) = 4f^{-1}(y) - 7 = 4\,\frac{y+7}{4} - 7 = y$$

and so $f \circ f^{-1}$ is also equal to ι.

Theorem. *Composition with inverse.* If $f : X \to Y$ is a bijection then

$$f^{-1} \circ f = \iota_X \quad \text{and} \quad f \circ f^{-1} = \iota_Y \ ,$$

where ι_X is the identity function on X and ι_Y is the identity function on Y.

Proof. Recall that

$$f^{-1}(y) = x \qquad \Leftrightarrow \qquad y = f(x) \ . \qquad (*)$$

In order to prove $f^{-1} \circ f = \iota_X$, we need to show that for every $x \in X$ the equation $f^{-1}\big(f(x)\big) = x$ holds.
Let $x \in X$.
Let $y = f(x)$.
Then, from $(*)$, $f^{-1}(y) = x, \ldots$
$\ldots$ and substituting $y = f(x)$ into this gives

$$f^{-1}\big(f(x)\big) = x \ .$$

Hence $f^{-1} \circ f = \iota_X$.

Exercise. Finish this proof by showing that for every $y \in Y$ the equation $f\big(f^{-1}(y)\big) = y$ holds.

Sometimes we need to find all results produced by applying a function to many elements, or all inputs which will give required function values.

Definition. Let f be a function from X to Y. If A is a subset of X, then $f(A)$ denotes the set of all values $f(x)$ with $x \in A$, that is,

$$f(A) = \{ f(x) \mid x \in A \} .$$

If $B \subseteq Y$, then $f^{-1}(B)$ denotes the set of all x which produce elements of B, that is,

$$f^{-1}(B) = \{ x \in X \mid f(x) \in B \} .$$

Examples.

1. Consider $f : \mathbb{R} \to \mathbb{R}$ where $f(x) = x^2$.
   - If $A = \{ x \in \mathbb{R} \mid 2 < x < 3 \}$ then
   $$\begin{aligned} f(A) &= \{ x^2 \mid 2 < x < 3 \} \\ &= \{ y \in \mathbb{R} \mid 4 < y < 9 \} . \end{aligned}$$
   - If $A = \{ x \in \mathbb{R} \mid -4 < x < 5 \}$ then
   $$\begin{aligned} f(A) &= \{ x^2 \mid -4 < x < 5 \} \\ &= \{ y \in \mathbb{R} \mid 0 \le y < 25 \} . \end{aligned}$$
   - If $B = \{ y \in \mathbb{R} \mid 0 \le y \le 9 \}$ then
   $$\begin{aligned} f^{-1}(B) &= \{ x \in \mathbb{R} \mid 0 \le x^2 \le 9 \} \\ &= \{ x \in \mathbb{R} \mid -3 \le x \le 3 \} . \end{aligned}$$
   - If $B = \{ y \in \mathbb{R} \mid 1 \le y < 16 \}$ then
   $$\begin{aligned} f^{-1}(B) &= \{ x \in \mathbb{R} \mid 1 \le x^2 < 16 \} \\ &= \{ x \in \mathbb{R} \mid -4 < x \le -1 \} \\ &\qquad \cup \{ x \in \mathbb{R} \mid 1 \le x < 4 \} . \end{aligned}$$
   - If $B = \{ y \in \mathbb{R} \mid y < -3 \}$ then
   $$f^{-1}(B) = \{ x \in \mathbb{R} \mid x^2 < -3 \} = \varnothing .$$

   Note that f is not an invertible function: if y is a real number, then $f^{-1}(y)$ does not exist. However f^{-1} of a set is different and will always exist (even if it's the empty set as in our last example). **Exercise**. What is $f^{-1}(\{ y \})$?

2. Define $f : \mathbb{N}^2 \to \mathbb{N}$ by $f(m,n) = mn$.
   - If $A = \{\, 0,1,2 \,\} \times \{\, 1,2,3,4 \,\}$ then
   $$f(A) = \{\, 0,1,2,3,4,6,8 \,\} .$$
   - If $B = \{\, 4,5,6 \,\}$ then
   $$f^{-1}(B) = \{\, (1,4),(2,2),(4,1),(1,5),(5,1),\\ (1,6),(2,3),(3,2),(6,1) \,\} .$$
3. Let $S = \{\, a,b,c,d \,\}$ and define a function f from $\mathcal{P}(S)$ to $\mathbb{N}$ by $f(X) = |X|$.
   - If $A = \{\, X \in \mathcal{P}(S) \mid a \in X \text{ and } d \in X \,\}$ then
   $$f(A) = \{\, 2,3,4 \,\} .$$
   - If $B = \{\, 1,4 \,\}$ then
   $$\begin{aligned} f^{-1}(B) &= \{\, X \in \mathcal{P}(S) \mid f(X) = 1 \text{ or } 4 \,\} \\ &= \{\, X \subseteq S \mid X \text{ has 1 or 4 elements} \,\} \\ &= \big\{\, \{\, a \,\},\{\, b \,\},\{\, c \,\},\{\, d \,\},\{\, a,b,c,d \,\} \,\big\} . \end{aligned}$$

SEQUENCES AND SUMMATIONS

Definition. A **sequence** is a function with domain a subset of $\mathbb{Z}$.

Notation. When discussing a sequence a we usually write a_n instead of $a(n)$. The sequence as a whole will be denoted by $\{ a_n \}$, or by the list

$$a_1, a_2, a_3, \dots \ .$$

Note.

- The domain of a sequence will usually be $\mathbb{N}$ or $\mathbb{Z}^+$, sometimes a finite set such as $\{ 1, 2, \dots, n \}$.
- The notation clashes with set notation, but it will always be clear from the context whether we are talking about a set or a sequence.
- Order and repetition of elements **are** important in sequences.

Examples.

1. If the function a is defined by $a_n = a(n) = 1/n$, then the sequence $\{ a_n \}$ with domain $\mathbb{Z}^+$ is

$$1,\ \tfrac{1}{2},\ \tfrac{1}{3},\ \tfrac{1}{4},\ \dots\ .$$

2. The sequence of primes, $\{ p_n \}$, is

$$2, 3, 5, 7, 11, 13, 17, \dots, 2^{82589933} - 1, \dots\ .$$

3. The Fibonacci numbers are the sequence $\{ f_n \}$ with domain $\mathbb{N}$:

$$\{ f_n \} = 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots\ .$$

4. If d_n is the nth digit of π, then d is a sequence with domain $\mathbb{N}$, thus:

$$\{ d_n \}_{n=0}^{\infty} = \{ 3, 1, 4, 1, 5, 9, 2, 6, 5, \dots \}\ .$$

5. A finite sequence is often called a **string** and written without commas. For example,

$$1100001\ , \quad 1001100\ , \quad 0111000$$

are three different **bit strings** of **length** 7.

Summation notation. The notation

$$\sum_{j=m}^{n} a_j \ ,$$

where $\{ a_j \}$ is a sequence and $m \leq n$, stands for

$$a_m + a_{m+1} + a_{m+2} + \cdots + a_n \ .$$

The **index of summation** j may be replaced by a different variable.

Examples.

1. $\displaystyle\sum_{i=1}^{5} i^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55.$

2. If $\{ f_k \}$ is the Fibonacci sequence,

$$\sum_{k=0}^{6} f_k = f_0 + f_1 + \cdots + f_6 = 20$$

and

$$\sum_{k=0}^{7} f_k = \left(\sum_{k=0}^{6} f_k \right) + f_7 = 20 + 13 = 33 \ .$$

Exercise. See if you can make a guess about the value of

$$\sum_{k=0}^{n} f_k \ .$$

Try to prove what you have guessed.

Exercise. Write in summation notation

$$\cos 1 + \cos 3 + \cos 5 + \cdots + \cos(2n-1)\ .$$

Answer. $\displaystyle\sum_{k=1}^{n} \cos(2k-1).$

Some basic sums. The following sums are often important.

$$\sum_{j=0}^{n} ar^j = a\,\frac{r^{n+1}-1}{r-1} \quad \text{provided } r \neq 1.$$

$$\sum_{j=1}^{n} 1 = \underbrace{1+1+1+\cdots+1}_{n \text{ times}} = n\ .$$

$$\sum_{j=1}^{n} j = \tfrac{1}{2}n(n+1)\ .$$

$$\sum_{j=1}^{n} j^2 = \tfrac{1}{6}n(n+1)(2n+1)\ .$$

Beware of a common error:

$$\sum_{j=0}^{n} 1 = \underbrace{1+1+1+\cdots+1}_{n+1 \text{ times}}\ ,$$

and this is equal to $n+1$, not n.

Transformation of sums. We can often work out more complicated sums by breaking them up into the basic sums just mentioned.

Addition of sums, multiplication by a constant. Let $\{ a_k \}$ and $\{ b_k \}$ be sequences. Then

$$\sum_{k=1}^{n}(a_k + b_k) = \Big(\sum_{k=1}^{n} a_k\Big) + \Big(\sum_{k=1}^{n} b_k\Big) \ ,$$

and if c is a constant,

$$\sum_{k=1}^{n} c\, a_k = c\Big(\sum_{k=1}^{n} a_k\Big) \ .$$

Example. Using the above properties,

$$\begin{aligned}
\sum_{k=1}^{12}(2k-5)^2 &= \sum_{k=1}^{12}(4k^2 - 20k + 25) \\
&= \sum_{k=1}^{12} 4k^2 - \sum_{k=1}^{12} 20k + \sum_{k=1}^{12} 25 \\
&= 4\sum_{k=1}^{12} k^2 - 20\sum_{k=1}^{12} k + 25\sum_{k=1}^{12} 1 \\
&= 4 \times \tfrac{1}{6} \times 12 \times 13 \times 25 \\
&\qquad - 20 \times \tfrac{1}{2} \times 12 \times 13 + 25 \times 12 \\
&= 2600 - 1560 + 300 \\
&= 1340 \ .
\end{aligned}$$

Shifting the index of summation is another useful technique. By substituting $k = j + p$ we have

$$\sum_{j=m}^{n} a_j = \sum_{k=m+p}^{n+p} a_{k-p} \ .$$

Example. Find the sum of all the squares from 100^2 to 200^2.
Solution. The sum is

$$\sum_{j=100}^{200} j^2 = \sum_{k=0}^{100} (k+100)^2$$

by substituting $k = j - 100$. Hence

$$\begin{aligned}\sum_{j=100}^{200} j^2 &= \sum_{k=0}^{100} k^2 + 200 \sum_{k=0}^{100} k + 10000 \sum_{k=0}^{100} 1 \\ &= \tfrac{1}{6} \times 100 \times 101 \times 201 \\ &\quad + 200 \times \tfrac{1}{2} \times 100 \times 101 \\ &\quad + 10000 \times 101 \\ &= 2358350 \ .\end{aligned}$$

Splitting the sum into two parts will sometimes achieve the same result:

$$\begin{aligned}\sum_{j=100}^{200} j^2 &= \sum_{j=1}^{200} j^2 - \sum_{j=1}^{99} j^2 \\ &= \tfrac{1}{6} \times 200 \times 201 \times 401 - \tfrac{1}{6} \times 99 \times 100 \times 199 \\ &= 2358350 \ .\end{aligned}$$

A particular kind of substitution is to **reverse the order** of the terms in the series:

$$\begin{aligned}\sum_{j=1}^{n} a_j &= a_1 + a_2 + \cdots + a_n \\ &= a_n + a_{n-1} + \cdots + a_1 \\ &= \sum_{k=1}^{n} a_{n+1-k} \ .\end{aligned}$$

We may think of this as making the substitution $k = n+1-j$. Similarly, we have

$$\sum_{j=0}^{n} a_j = \sum_{k=0}^{n} a_{n-k}$$

by setting $k = n - j$.

Exercise. Use an appropriate substitution to show that

$$\sum_{j=m}^{n} a_j = \sum_{k=m}^{n} a_{m+n-k} \ .$$

Example. This technique is useful in proving the formula for the sum of an arithmetic progression:

$$\sum_{k=1}^{n}\bigl(a+(k-1)d\bigr)=\tfrac{1}{2}n\bigl(2a+(n-1)d\bigr)\ .$$

Proof. Let $S = \text{LHS}$. Substituting $j = n + 1 - k$, we have

$$S=\sum_{j=1}^{n}\bigl(a+(n-j)d\bigr)=\sum_{k=1}^{n}\bigl(a+(n-k)d\bigr)\ .$$

Hence

$$\begin{aligned}2S&=\sum_{k=1}^{n}\bigl(a+(k-1)d\bigr)+\sum_{k=1}^{n}\bigl(a+(n-k)d\bigr)\\&=\sum_{k=1}^{n}\bigl(2a+(n-1)d\bigr)\\&=\bigl(2a+(n-1)d\bigr)\sum_{k=1}^{n}1\\&=n\bigl(2a+(n-1)d\bigr)\end{aligned}$$

and so

$$S=\tfrac{1}{2}n\bigl(2a+(n-1)d\bigr)\ .$$

Telescoping sums are those which may be written out in such a way that most of the terms cancel and the rest are easy to add up.

Examples.

1. We have

$$\begin{aligned}
&\sum_{k=1}^{n}\big(\sin(2k)-\sin(2k-2)\big)\\
&\qquad=\sum_{k=1}^{n}\sin(2k)-\sum_{k=1}^{n}\sin(2k-2)\\
&\qquad=\big(\sin 2+\sin 4+\cdots+\sin(2n-2)+\sin(2n)\big)\\
&\qquad\qquad-\big(\sin 0+\sin 2+\cdots+\sin(2n-4)+\sin(2n-2)\big)\\
&\qquad=\sin 2n-\sin 0\\
&\qquad=\sin 2n\ .
\end{aligned}$$

2. Similarly

$$
\begin{aligned}
\sum_{k=1}^{n}\Big(\frac{1}{k}-\frac{1}{k+2}\Big)
&= \sum_{k=1}^{n}\frac{1}{k}-\sum_{k=1}^{n}\frac{1}{k+2}\\
&= \left(1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots+\frac{1}{n}\right)\\
&\qquad -\left(\frac{1}{3}+\frac{1}{4}+\cdots+\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}\right)\\
&= 1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2}\ .
\end{aligned}
$$

Alternatively, we can set this out using a shift of summation index:

$$
\begin{aligned}
\sum_{k=1}^{n}\Big(\frac{1}{k}-\frac{1}{k+2}\Big) &= \sum_{k=1}^{n}\frac{1}{k}-\sum_{k=1}^{n}\frac{1}{k+2}\\
&= \sum_{k=1}^{n}\frac{1}{k}-\sum_{j=3}^{n+2}\frac{1}{j}\\
&= \left(1+\frac{1}{2}+\sum_{k=3}^{n}\frac{1}{k}\right)\\
&\qquad -\Big(\sum_{j=3}^{n}\frac{1}{j}+\frac{1}{n+1}+\frac{1}{n+2}\Big)\\
&= 1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2}\ .
\end{aligned}
$$

Of course, the sums we have just seen are easy because we have made them easy! However other sums, which may not at first seem to be of this form, can sometimes be written as telescoping sums.

Examples.

1. Find $\displaystyle\sum_{k=1}^{n} \cos(2k-1)$.

   Solution. The products–to–sums formula

   $$2 \sin 1 \cos(2k-1) = \sin(2k) - \sin(2k-2)$$

   can be used to obtain

   $$\sum_{k=1}^{n} \cos(2k-1) = \frac{1}{2\sin 1} \sum_{k=1}^{n} \sin(2k) - \sin(2k-2) = \frac{\sin 2n}{2 \sin 1} \, .$$

2. Find $\displaystyle\sum_{k=1}^{10} \frac{2}{k^2+2k}$.

   Solution. By the method of partial fractions,

   $$\frac{2}{k^2+2k} = \frac{2}{k(k+2)} = \frac{1}{k} - \frac{1}{k+2} \, .$$

   Hence, from the previous slide,

   $$\sum_{k=1}^{10} \frac{2}{k^2+2k} = 1 + \frac{1}{2} - \frac{1}{11} - \frac{1}{12} = \frac{175}{132} \, .$$

3. A harder example: find $\displaystyle\sum_{k=2}^{n} \frac{2}{k^3 - k}$.

Solution. We have

$$\frac{2}{k^3 - k} = \frac{2}{(k-1)k(k+1)} = \frac{1}{k-1} - \frac{2}{k} + \frac{1}{k+1} \ ,$$

so

$$\begin{aligned}
\sum_{k=2}^{n} \frac{2}{k^3 - k} &= \sum_{k=2}^{n} \frac{1}{k-1} - 2\sum_{k=2}^{n} \frac{1}{k} + \sum_{k=2}^{n} \frac{1}{k+1} \\
&= \Big(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n-1}\Big) \\
&\qquad - 2\Big(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n-1} + \frac{1}{n}\Big) \\
&\qquad\qquad + \Big(\frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n-1} + \frac{1}{n} + \frac{1}{n+1}\Big) \\
&= 1 + \frac{1}{2} - \frac{2}{2} - \frac{2}{n} + \frac{1}{n} + \frac{1}{n+1} \\
&= \frac{1}{2} - \frac{1}{n} + \frac{1}{n+1} \\
&= \frac{1}{2} - \frac{1}{n^2 + n} \ .
\end{aligned}$$

4. Simplify $k^2(k+1)^2-(k-1)^2k^2$; hence evaluate $\displaystyle\sum_{k=1}^{n} k^3$.

Solution. We have

$$\begin{aligned} k^2(k+1)^2-(k-1)^2k^2 &= k^2\big((k+1)^2-(k-1)^2\big) \\ &= 4k^3 \end{aligned}$$

and so

$$\begin{aligned} \sum_{k=1}^{n} k^3 &= \frac{1}{4}\sum_{k=1}^{n} 4k^3 \\ &= \frac{1}{4}\sum_{k=1}^{n}\big(k^2(k+1)^2-(k-1)^2k^2\big) \\ &= \frac{1}{4}\Big[\big(1^2\times 2^2+\cdots+(n-1)^2n^2+n^2(n+1)^2\big) \\ &\qquad\qquad -\big(0^2\times 1^2+1^2\times 2^2+\cdots+(n-1)^2n^2\big)\Big] \\ &= \frac{n^2(n+1)^2}{4} \\ &= \left(\frac{n(n+1)}{2}\right)^2 . \end{aligned}$$